

upstream of Group II *nif*-gene
upstream of Group I *nif*-gene

bits

1 10 20 30 40 50 60 70 80 89

a dg5 *nifB* upstream -621
a dg5 *nifP* upstream -866
M55-K2018 *nifB* upstream -280
M55-K2018 *nifP* upstream -230
braeum IMS101 *nifB* upstream -344
braeum IMS101 *nifP* upstream -856
C7327 *nifP* upstream -235
C7327 *nifB* upstream -246
is NIES-593 *nifB1* upstream -204
is NIES-593 *nifP1* upstream -458
Chonoplastes PCC7420 *nifB* upstream -394
Chonoplastes PCC7420 *nifP* upstream -230
A1-SPW-01 *nifX* upstream -398
A1-SPW-01 *nifP* upstream -230
is ATCC 29413 *nifB1* upstream -538
is ATCC 29413 *nifP1* upstream -232
is ATCC 29413 *nifB2* upstream -480
C09 *nifB1* upstream -932
C09 *nifX2* upstream -294
C09 *nifP* upstream -360
CY1219 *nifX* upstream -479

GAGTAC-AGAAAAACCAACCCCGCC-CCTTAACTCAATCCCTCTGTAAAAAACTCGCACCCCTGCACCCCAAAATAATTTT
GAGTAC-AGAAAAACCAACCCCGCC-CCTTAACTCAATCCCTCTGTAAAAAACTCGCACCCCTGCACCCCAAAATAATTTT
GAGTAC-AGGAATACAGATCCCTTCTCTACAAATACTTGCACCTCTTAAGAAAAACTGTCTCTGCACCCGAAATAAGATTTT
GAGTGC-AGAAAGCTCAGTCCCCAT-ATAAATACTCCCAACCCGTTGTAAAAAACTAGTCCCTGCACCCAAATAGTTTTT
GAGTACATACACTACTGTACCATTACC-CCATAAATACATAATACCCCACTAAAAAAACAAGTACCTATACCCCAAAAAAAATTTT
GAGTACATACACTACTGTACCATTACC-CCATAAATACATAATACCCCACTAAAAAAACAAGTACCTATACCCCAAAAAAAATTTT
GAGTAC-ACAAATATCTGCGCCTCCAT-GCATAAATACTCTAA-CCCTATTAAAAAAACTAGCCCTCTGACCCAAAAAATTTTTT
GAGTAC-ACAAATACCGTCCCTTC-T-ACAAAAAACTAGCTCTCTCTCCCAAAAAAACTAGCCCTCAACCTCAAAAAATTTTTT
GAGTAC-ACAAATATCTGCGCCTCCAT-TCATAAATACTCTAAACCCCTATTAAAAAAACTAGCCCTTGGCCCAAAAAATTTTTT
GAGTAC-AAAACTCTTAACGCCCCACC-CATTAAATACTTGCACCCCATCCGGAATACTAGAACCCCTACCCCAAGAAATTTTTT
GAGTAC-ACAAATCTAGAACCTCGAC-CCGAAAACTTGGCATCCGGGATAAAAAAACGAGAACCTTATCCCC-----
GAGTAC-AAAACTCTTAACGCCCCACC-CATTAAATACTTGCACCCCATCCGGAATACTAGAACCCCTACCCCAAGAAATTTTTT
GAGTAC-ACAAATCTAGAACCTCGAC-CCAAAACTTGGCATCCGGGATAAAAAAACGAGAACCCCGATCCCC-----
-AGTACAATCAGCTCTGAAACCCCTTT-CCATAAATACTCGTACCCCAACTAGAAAAAACTAGAACCCCTACCTGGAAAAAATTTTTT
GAGTAC-AAAAATTTCTGAACCCCTCGT-CTAAAAAACTAGTACCCCTGCACAAAAAACTAGTACCTATATCCCAAGAGAAATTTTTT
-AGTAC-ATAAATACTTACACCCCTTCATCATAAATACTCGTACCCCTGGCAGAAAAAACTAGCACCCATACCTAAAAAATTTTTT
GAGTGC-AAAAACCCGCAACCCCGGAC-TGAAAAACACCGTCCCCCCCCGCCCAAAATACCCGCCCCCCACCCCAAAAAATTTTTT
GAGTGCAGTTTCTTCCGTCCTCCCGCC-TCAAAAAAGTCTGTCCCCCCCCCAAAAGTCCCGAACCCCGATCCCGATCGCAAAATTTTTT
GAGTGC-AAAAACCCGCAACCCCGGAC-TGAAAAACACCGTCCCCCCCCGCCCAAAATACCCGCCCCCCACCCCAAAAAATTTTTT
GAGTAC-AAAAAAGCCGCAACCCACCT-CCGCAAAATACCGTACCTCCCCGGAGAAACGCCCGACCCCGGAGGCTGAAAAAAATTAATTTT

B

Phylogenetic tree showing the relationships between various CnfR proteins. The tree is rooted at the center and branches outwards. Bootstrap values are indicated at the nodes. The tree is divided into three main clades: CnfR Clade I (red), CnfR Clade II (blue), and CnfR Clade III (dark blue). The legend indicates that pink squares represent the presence of the N-terminal 4Fe-4S motif and orange squares represent the presence of the C-terminal HTH motif.

CnfR Clade I

- 1. *L. boryana* dg5 CnfR
- 2. *S. M55-k2018* CnfR
- 3. *T. erythreum* IMS101 CnfR
- 4. *P. sp.* PCC 7327 CnfR1
- 5. *H. rivularis* NIES 593 CnfR1
- 6. *C. chloroplastes* PCC 7420 CnfR
- 7. *C. sp.* AI-SPW-01 CnfR
- 8. *T. variabilis* ATCC 29413 CnfR2
- 9. *L. sp.* CCY1209 CnfR1

CnfR Clade II

- 10. *P. sp.* CCY1219 CnfR
- 11. *H. rivularis* NIES 593 CnfR2
- 12. *P. sp.* PCC 7327 CnfR2
- 13. *P. sp.* LEGE 06147 CnfR
- 14. *G. sp.* PCC 9228 CnfR1
- 15. *B. sp.* CnfR like
- 16. *L. sp.* CCY1209 CnfR2

CnfR Clade III

- 17. *P. sp.* PBR-2020 CnfR
- 18. *S. sp.* AB48 CnfR

Tree scale: 1

Legend:

- Presence of N-terminal 4Fe-4S motif
- Presence of C-terminal HTH motif

Cnfr protein

protein length

P. sp. PBR-2020 Cnfr 426

S. sp. AB48 Cnfr 427

P. sp. LEGE 06147 Cnfr 514

L. sp. CCY1209 Cnfr 516

Geitlerinema PCC 9228 Cnfr1 624

① *L. boryana* dg5 Cnfr 635

② *S. M55-k2018* Cnfr 544

⑧ *T. variabilis* ATCC 29413 Cnfr1 529

⑧ *T. variabilis* ATCC 29413 Cnfr2 531

⑨ *L. sp.* CCY1209 Cnfr1 531

⑤ *H. rivularis* NIES 593 Cnfr1 533

④ *P. sp.* PCC 7327 Cnfr1 533

③ *T. erythraeum* IMS101 Cnfr 530

⑦ *C. sp.* A1-SPW-01 Cnfr 537

⑥ *C. chthonoplastes* PCC 7420 Cnfr 537

Baaleninema sp. Cnfr like 237

H. rivularis NIES 593 Cnfr2 539

⑩ *Phormidium* sp. CCY1219 Cnfr 472

P. sp. PBR-2020 Cnfr

S. sp. AB48 Cnfr

P. sp. LEGE 06147 Cnfr

L. sp. CCY1209 Cnfr2

Geitlerinema PCC 9228 Cnfr1

① *L. boryana* dg5 Cnfr

② *S. M55-k2018* Cnfr

⑧ *T. variabilis* ATCC 29413 Cnfr1

⑧ *T. variabilis* ATCC 29413 Cnfr2

⑨ *L. sp.* CCY1209 Cnfr1

⑤ *H. rivularis* NIES 593 Cnfr1

④ *P. sp.* PCC 7327 Cnfr1

③ *T. erythraeum* IMS101 Cnfr

⑦ *C. sp.* A1-SPW-01 Cnfr

⑥ *C. chthonoplastes* PCC 7420 Cnfr

Baaleninema sp. Cnfr like

H. rivularis NIES 593 Cnfr2

⑩ *Phormidium* sp. CCY1219 Cnfr

Pairwise identity (%)

70

60

50

40

35